

Python Portfolio — Tech Stack and Dependencies

Project: Python Portfolio
Technology: Python, FastAPI, SQLAlchemy, SQLite, Pydantic, JWT
Location: /home/dominic/Documents/python-portfolio
Report #: 5/20

Primary Stack

Python, FastAPI, SQLAlchemy, SQLite, Pydantic, JWT

Key Dependencies

- fastapi — API framework
 - sqlalchemy — ORM
 - aiosqlite — async SQLite
 - python-jose — JWT
 - passlib — password hashing
 - httpx — HTTP client (chat module)
 - openai — LLM client (RAG/chat)
 - chromadb — vector DB (RAG)
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Technology Stack Details

Framework Versions

Component	Technology	Version	Purpose
Backend	FastAPI	API server	
Database	SQLite	Persistence	
Auth	JWT	Security	
Frontend	None	UI	
Hosting	Render.com + GitHub Pages	Deployment	

Core Dependencies

```
# Core dependencies
fastapi>=0.115.0
sqlalchemy>=2.0.0
pydantic>=2.0.0
uvicorn[standard]>=0.30.0
```

Dependency Justification

Package	Why	Alternative Considered
fastapi	Modern ASGI framework	Flask, Django
sqlalchemy	Mature ORM with async	peewee, tortoise-orm
pydantic	Validation + serialization	marshmallow
jose	JWT handling	PyJWT

Development Dependencies

Package	Purpose
pytest	Testing framework
httpx	Async HTTP client for tests
uvicorn	ASGI server
python-dotenv	Environment variable loading

External Services

Service	Purpose	Integration Point
OpenAI	AI chat completion	/v1/chat/completions proxy
GitHub Pages	Static hosting	git push deployment

Version Constraints

- Python 3.10+ required (type hints, pattern matching)
- Pydantic v2 for improved validation performance
- SQLAlchemy 2.0+ for modern async support